

Estimating Difference between Two Means.

Assume we have 2 populations

⇒ For POP 1:

$$\mu_1, N_1, \sigma_1^2$$

⇒ For POP 2:

$$\mu_2, N_2, \sigma_2^2$$

Now, we computed

$\bar{X}_1, \bar{X}_2, S_1^2, S_2^2$ for both POP.

Hypothesis:

$$H_0 : \mu_1 - \mu_2 = d_0 \quad] \text{ Null}$$

$$H_1 : \mu_1 - \mu_2 \neq d_0$$

$$H_1 : \mu_1 - \mu_2 > d_0$$

$$H_1 : \mu_1 - \mu_2 < d_0$$

} Alternative

1st Situation:

σ_1, σ_2 known.

(z-test is used in this situation)

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where d_0 is the claim.

Date: _____

Day: _____

2nd Situation:

$\sigma_1 = \sigma_2$ but unknown.

T-test is used in this situation.

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where $V = n_1 + n_2 - 2$

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}$$

Formula of S_p^2 will be given.

3rd Situation:

$\sigma_1 \neq \sigma_2$ but unknown

F-test is used

$$t' = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

where

$$V = \frac{(S_1^2/n_1 + S_2^2/n_2)^2}{\frac{(S_1^2/n_1)^2}{n_1 - 1} + \frac{(S_2^2/n_2)^2}{n_2 - 1}}$$

In this case formula of V will be given.

Question: Two-Sample t-Test (Equal Variances Assumed)

A researcher wants to determine whether the difference in mean scores between two groups **exceeds 2 units**. The following data is collected: **Group 1** Sample size (n_1) = 12 Sample mean ($\bar{x}_1$) = 85 Sample standard deviation (s_1) = 4 **Group 2** Sample size (n_2) = 10 Sample mean ($\bar{x}_2$) = 81 Sample standard deviation (s_2) = 5 Assuming that the populations **have equal variances**, test at the **0.05 significance level** whether the mean difference between the two groups **exceeds 2 units**.

Data:

$$\begin{aligned}N_1 &= 12, & N_2 &= 10 \\ \bar{x}_1 &= 85, & \bar{x}_2 &= 81 \\ s_1 &= 4, & s_2 &= 5 \\ \alpha &= 0.05.\end{aligned}$$

=> Hypothesis:

$$\begin{aligned}H_0: \mu_1 - \mu_2 &= 2 \\ H_1: \mu_1 - \mu_2 &> 2\end{aligned}$$

=> Test Statistics:

$$\Rightarrow t = \frac{(\bar{x}_1 - \bar{x}_2) - d_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad - (1)$$

$$\Rightarrow S_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(12 - 1)(16) + (10 - 1)(25)}{12 + 10 - 2}$$

$$S_p^2 = 20.05$$

$$S_p = 4.47$$

put in eq (1)

$$t = \frac{(85 - 81) - 2}{4.47 \sqrt{\frac{1}{12} + \frac{1}{10}}}$$

$$t_0 = 1.044$$

=> Critical Region:

$$V = N_1 + N_2 - 2$$
$$= 20$$

$$\alpha = 0.05$$

$$T_{0.05}(20) = 1.725$$

as $t_0 > t_{\alpha/2}(v)$

1.044 is not greater
than 1.725

so

we will accept H_0